EXPERIMENT 3: MINIMAX ALGORITHM

AIM

To use the MINIMAX algorithm to evaluate the optimal move in a game tree.

PROCEDURE

1. Define the terminal values at the leaf nodes.
2. Apply the recursive MINIMAX algorithm from the root toward the leaves.
3. Alternate between maximizing and minimizing levels to determine the optimal value.

CODE (PYTHON)

```
def minimax(depth, nodeIndex, isMax, scores, h):
    if depth == h:
        return scores[nodeIndex]

    if isMax:
        return max(
            minimax(depth + 1, nodeIndex * 2, False, scores, h),
            minimax(depth + 1, nodeIndex * 2 + 1, False, scores, h)
        )
    else:
        return min(
            minimax(depth + 1, nodeIndex * 2, True, scores, h),
            minimax(depth + 1, nodeIndex * 2 + 1, True, scores, h)
        )

scores = [3, 5, 6, 9, 1, 2, 0, -1]
h = 3
print("Optimal value:", minimax(0, 0, True, scores, h))
```

OUTPUT

```
Optimal value: 5
```

RESULT

Thus, the MINIMAX algorithm was implemented successfully to evaluate the optimal value in a game tree.